

MAT 426, Class 24 Homework: Orthogonal Sets
(Due Thursday, May 7, 2026)

1. Determine the unit circle in $\mathbb{R}^2$ for the given inner product.

(a) $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + \frac{1}{4}u_2v_2$

(b) $\langle \mathbf{u}, \mathbf{v} \rangle = 4u_1v_1 + u_1v_2 + u_2v_1 + 4u_2v_2$

2. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V . Prove that $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

3. Let $V = \mathbb{R}^2$. Consider the basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ for V . Find an orthogonal basis for V with respect to the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$.

4. Let $\mathcal{P}_2$ be the vector space of all polynomials of degree at most two. Consider the basis $\mathcal{B} = \{1, 1+x, 1+x+x^2\}$. Find an orthogonal basis for V with respect to the inner product

$$\langle f(x), g(x) \rangle = a_0b_0 + a_1b_1 + a_2b_2$$

where $f(x) = a_0 + a_1x + a_2x^2$ and $g(x) = b_0 + b_1x + b_2x^2$.

5. Let $\mathcal{P}_2[0, 1]$ be the vector space of all polynomials of degree at most two on the interval $[0, 1]$. Consider the basis $\mathcal{B} = \{1, 1+x, 1+x+x^2\}$. Find an orthogonal basis for V with respect to the inner product $\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x) dx$.

6. Let $\mathcal{P}$ be the vector space of all polynomials on the interval $(-\infty, \infty)$. Consider the weighted inner product

$$\langle f(x), g(x) \rangle = \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2} dx.$$

Starting with the standard basis $\{1, x, x^2, x^3, \dots\}$ construct the first four orthogonal polynomials in the orthogonal basis $\{H_0(x), H_1(x), H_2(x), H_3(x), \dots\}$ with respect to this inner product.